Data Structures: Introspection

For the task of week 4, I utilized a Binary Search over a Quicksearch. I chose to do a Binary Search because I found it to be quite convenient as well as more challenging and intriguing. The Binary Search took 4ms while the Breadth First Search took 2782ms, which does align with the expected results considering that a Binary Search is quite fast.

When performing the final week's assignment, I performed a Binary Search Tree which visited the elements of the tree through a Breadth First Search. While I at first intended to do a Hash Map, I decided to do a Binary Search Tree instead of a Hash Map since I felt like I understood it better than Hash Maps at the start of week 5. I also made two different scripts for the assignment, one with the tree and its node, and one with the Task itself.

While I was able to understand how things worked in theory during the lectures, two reoccurring issues that I had throughout the course were where to start (what scripts to create to begin the assignment) and that I had a hard time imagining how to write the code for the methods we were to implement. However, I was able to overcome these obstacles by being in school and asking the Lecturer and my classmates in-person. Another thing that helped me was when I realized during the Sorting Task of week 3 that each of the assignments require us to create a Task, which makes them slightly similar to each other.

Some things I needed were not in the lectures, and while I heard it was very easy to find on my own on the internet, I got some help from my classmates with these problems. Those problems included measuring the time for the Binary Search-Task and the Binary Search Tree-Task, and how to convert the data from the Json-file.

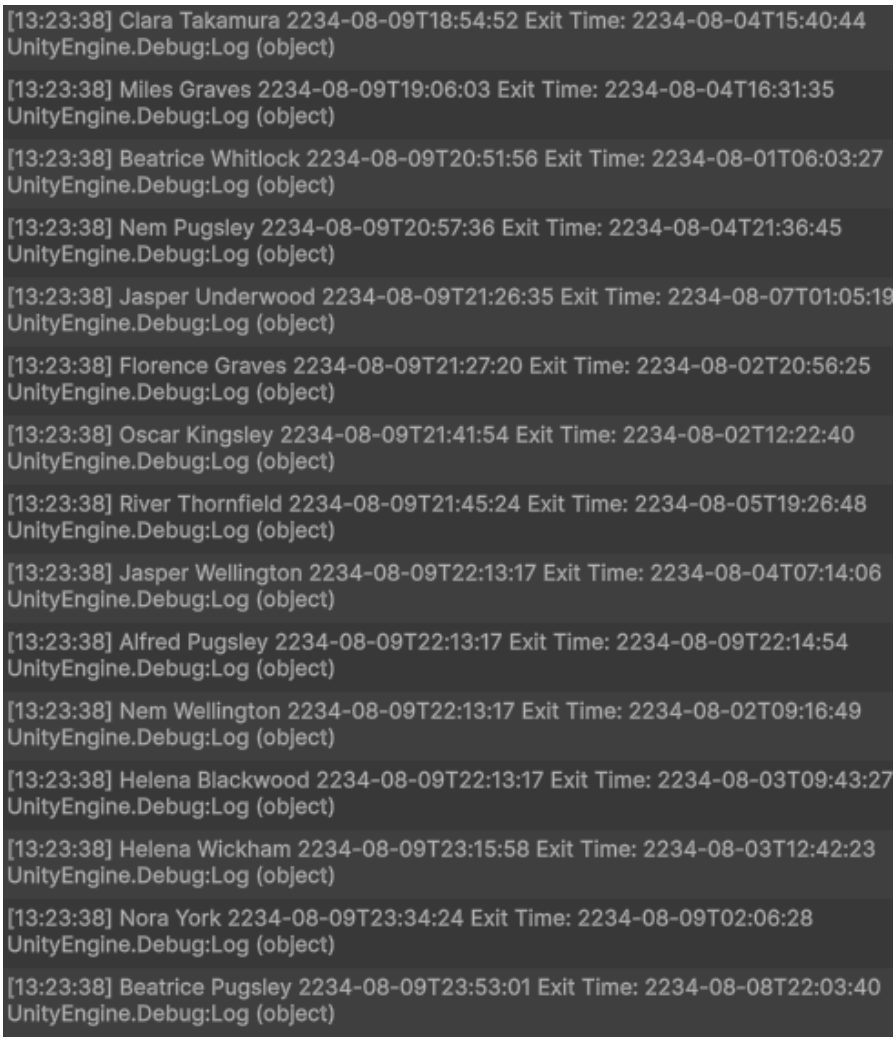


Figure 1 : The range of suspects that was found through the search.